PHY202: Physics II: Exam 6
Dr. Deskins

Excellent!

$\frac{9}{9} \Rightarrow \underline{100}\%$

0	6	11	17	22	28	33	39	44	50	56	61	67	72	78	83	89	94	100
0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5	9

Name:

These questions assess your understanding of the material from Chapters 25, 26 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. What power of spectacle lens is needed to correct the vision of a near-sighted person whose far point is 100 cm? Assume the spectacle lens is held 1.1 cm away from the eye by eyeglass frames.

ANSWER: -1.011 D ✓

$$P = \left(\frac{1}{d_o} + \frac{1}{d_i}\right)$$

$d_o = \infty$

$d_i = (1.1\,cm - 100\,cm)$

$d_i = -0.989\,m$

$$P = \left(\frac{1}{\infty} - \frac{1}{0.989}\right)$$

$$P = -1.011\,D$$

2. A clear-glass light bulb is placed 1.1 m from a thin convex lens that has a focal length of 0.096 m. Calculate the magnification of the produced image.

ANSWER: -0.0956 ✓

$$m = \frac{-d_i}{d_o}$$

$$d_i = \left(\frac{1}{f} - \frac{1}{d_o}\right)^{-1}$$

$$d_i = \left(\frac{1}{0.096} - \frac{1}{1.1}\right)^{-1} = 0.105179\,m$$

$$m = \frac{-0.105179}{1.1} = -0.0956$$

3. Suppose that you would like to find the angle of refraction in a material. You shine a laser through air at an incident angle of 2.1° into the material which has an index of refraction of 1.5. Calculate the angle of refraction in the material.

ANSWER: 1.4° from normal ✓

$$n_1 \sin 2.1° = n_2 \sin \theta_2$$

$n_1 = 1$

$n_2 = 1.5$

$$\sin^{-1}\left(\frac{\sin(2.1)}{1.5}\right) = \theta_2 = 1.3998° \approx 1.4°$$

+3

4. Suppose that you take a magnifying glass out on a sunny day and you find that it concentrates sunlight to a small spot 3.9 cm away from the lens. What is the power of the lens?

$\frac{1}{f} = P = \frac{1}{0.039} = 25.641$

ANSWER: 25.641 D ✓

5. You shine a laser through air at an incident angle of 25.° into an unknown material and measure the angle of refraction to be 34.°. What is the unknown material's index of refraction?

$\cancel{n_1} \sin 25° = n_2 \sin 34°$ $n_1 = 1$, $n_2 = ?$

$\Rightarrow \frac{\sin 25}{\sin 34} = n_2 = 0.75576$

ANSWER: 0.756 ✓

6. Calculate the power of the eye when viewing an object 18. cm away. Take the lens-to-retina distance to be 1.9 cm.

$P = \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{0.18} + \frac{1}{0.019}$

$= 58.187$ D

ANSWER: 58.19 D ✓

7. What is the size of the image on the retina of a 2.3 mm-diameter strand of hair that is held at arm's length (67. cm) away? Take the lens-to-retina distance to be 2.0 cm.

$\frac{h_i}{h_o} = \frac{-d_i}{d_o} \Rightarrow h_i = h_o \times \frac{-d_i}{d_o}$

$h_i = (2.3 \times 10^{-3}) \times \frac{-0.02}{0.67} = -6.866 \times 10^{-5}$

ANSWER: -6.866×10^{-5} D ✓

+4

8. A clear-glass light bulb is placed 0.62 m from a thin convex lens that has a focal length of 0.47 m. Calculate the distance from the lens to the location of the produced image.

$$d_i = \left(\frac{1}{f} - \frac{1}{d_o}\right)^{-1} = \left(\frac{1}{0.47} - \frac{1}{0.62}\right)^{-1}$$

ANSWER: 1.9427 m

✓

$$\Rightarrow d_i = 1.9427\ m$$

9. What power of spectacle lens is needed to correct the vision of a far-sighted person whose near point is 0.76 m, so that they may see an object clearly that is 11. cm away? Assume the spectacle lens is held 0.92 cm away from the eye by eyeglass frames.

$$d_o = (0.11 - 0.0092) = 0.1008$$

ANSWER: 8.5887 D

✓

$$d_i = (0.0092 - 0.76) = -0.7508$$

$$P = \left(\frac{1}{0.1008} - \frac{1}{0.7508}\right) = 8.5887\ D$$

+2

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1. $P = -1.01\,\mathrm{D} \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f},\ P = \frac{1}{f}$
2. $m = -0.0956 \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f},\ \frac{h_i}{h_0} = -\frac{d_i}{d_0} = m$
3. $\theta_2 = 1.40° \because n_1 \sin\theta_1 = n_2 \sin\theta_2$
4. $P = 25.6\,\mathrm{D} \because P = \frac{1}{f}$
5. $n_2 = 0.756 \because n_1 \sin\theta_1 = n_2 \sin\theta_2$
6. $P = 58.2\,\mathrm{D} \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f},\ P = \frac{1}{f}$
7. $h_i = -6.87 \times 10^{-5}\ \mathrm{m} \because \frac{h_i}{h_0} = -\frac{d_i}{d_0} = m$
8. $d_i = 1.94\,\mathrm{m} \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}$
9. $P = 8.59\,\mathrm{D} \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f},\ P = \frac{1}{f}$